Undercut effect

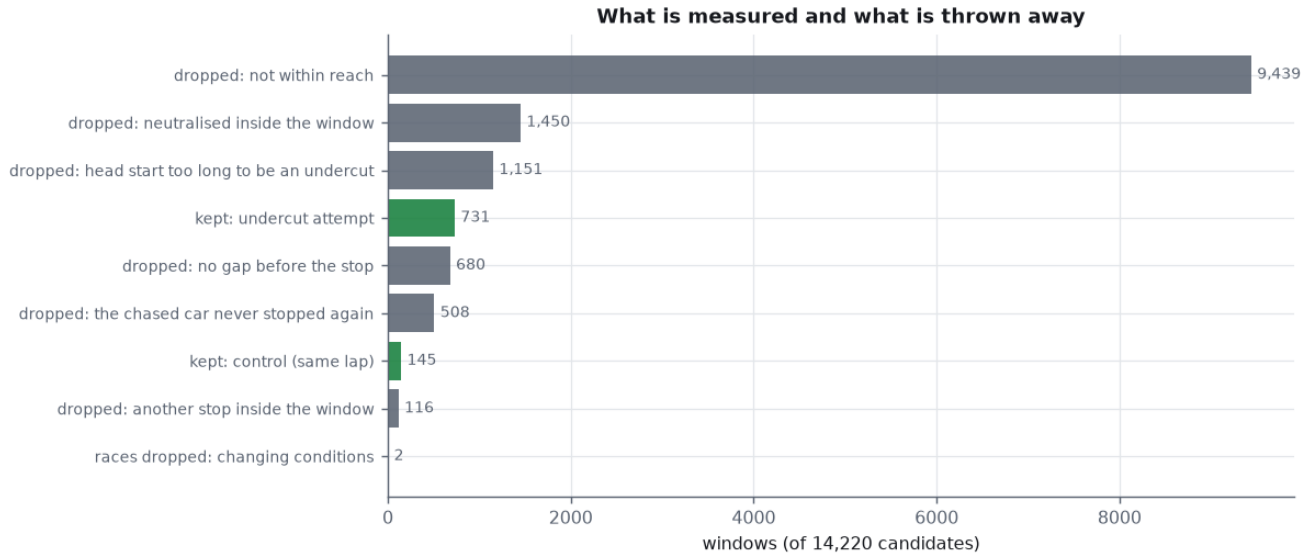
What pitting first is actually worth, measured from 2023 lap times alone — with a control group, because teams do not attempt undercuts at random.

Worth +3.82 s against a control, not against zero

Best at a head start of 3 laps, over 731 attempts against 145 same-lap controls. It clears the car 11% of the time.

731	145	+3.82 s	11%
undercut attempts	same-lap controls	best gain vs control	attempts that cleared the car

The gap is in the lap times. Summing a car's lap times gives its elapsed race time at the end of every lap, so the difference between two cars' sums is the gap between them. Exact, no interpolation, no telemetry. Everything here rests on that one observation.



Which windows are measured

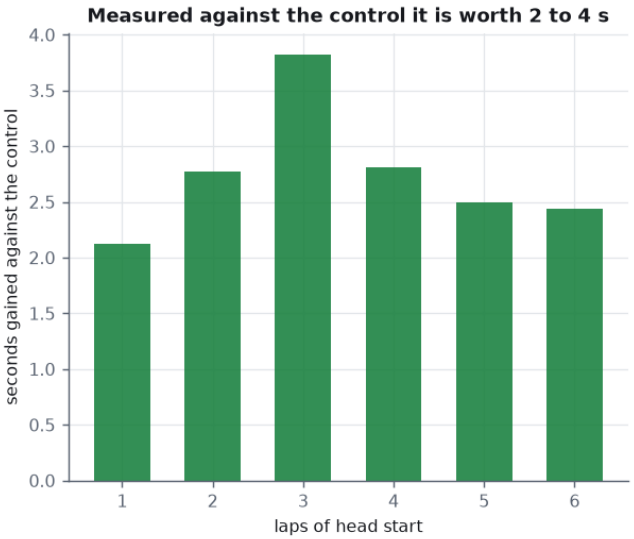
outcome	windows
candidate windows	14,220
dropped: not within reach	9,439
dropped: neutralised inside the window	1,450
dropped: head start too long to be an undercut	1,151
kept: undercut attempt	731
dropped: no gap before the stop	680
dropped: the chased car never stopped again	508
kept: control (same lap)	145
dropped: another stop inside the window	116
races dropped: changing conditions	2

In this schema a stop recorded on lap L puts 2.85 s of excess on lap L and 19.21 s on lap L+1 — the pit-lane time is mostly in the out-lap. So on the lap the chased car pits, the chasing car still reads -17.0 s: a pit loss not yet paid, not an undercut that failed. Every horizon below is counted from one lap after the second stop, the first point at which both cars have paid the same toll.

What it is worth

The control is not zero. A car that is behind is usually behind because it is slower, and in these windows it loses about two seconds to the car ahead even when neither gains any tyre advantage. Measured against zero the undercut looks worthless; measured against the control it is worth two to four seconds, best at a head start of 3 laps.





laps of head start	pairs	seconds closed (95%)	against control	rival tyre age
control (same lap)	145	-2.16 ± 0.81	+0.00	17
1	179	-0.04 ± 0.98	+2.12	16
2	156	+0.61 ± 1.18	+2.77	16
3	118	+1.66 ± 1.35	+3.82	16
4	104	+0.64 ± 1.41	+2.81	14
5	96	+0.33 ± 1.41	+2.49	15
6	77	+0.27 ± 1.58	+2.44	14

Does the gain run through tyre age?

The pure-pace model says the whole gain is the rival's tyre age times the head start, with no free term for the head start on its own. Fitting both terms tests it.

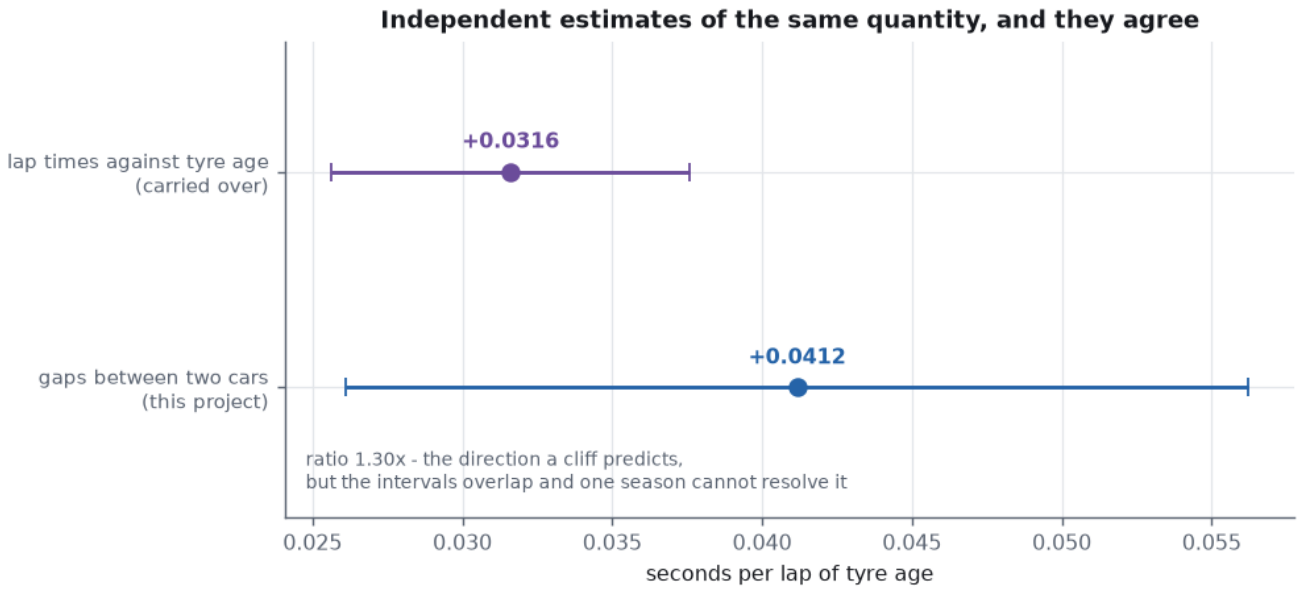
The free head-start term is indistinguishable from zero, so the form the physics implies is not rejected and the gain can be pushed entirely through the age channel — which turns its coefficient into an estimate of the tyre-age penalty itself.

term	estimate	95% interval	p
control	-0.7400	-1.8916 to +0.4117	0.2066
per lap of head start	+0.1237	-0.8201 to +1.0676	0.7963
per lap of head start x age	+0.0364	-0.0018 to +0.0745	0.0618
prior pace	-0.3497	-0.5855 to -0.1139	0.0038

A second opinion on tyre degradation

Two estimators of the same quantity, built from different features of the same season: one regresses a car's lap times on its own tyre age, the other reads the pace difference between two cars over a handful of laps.

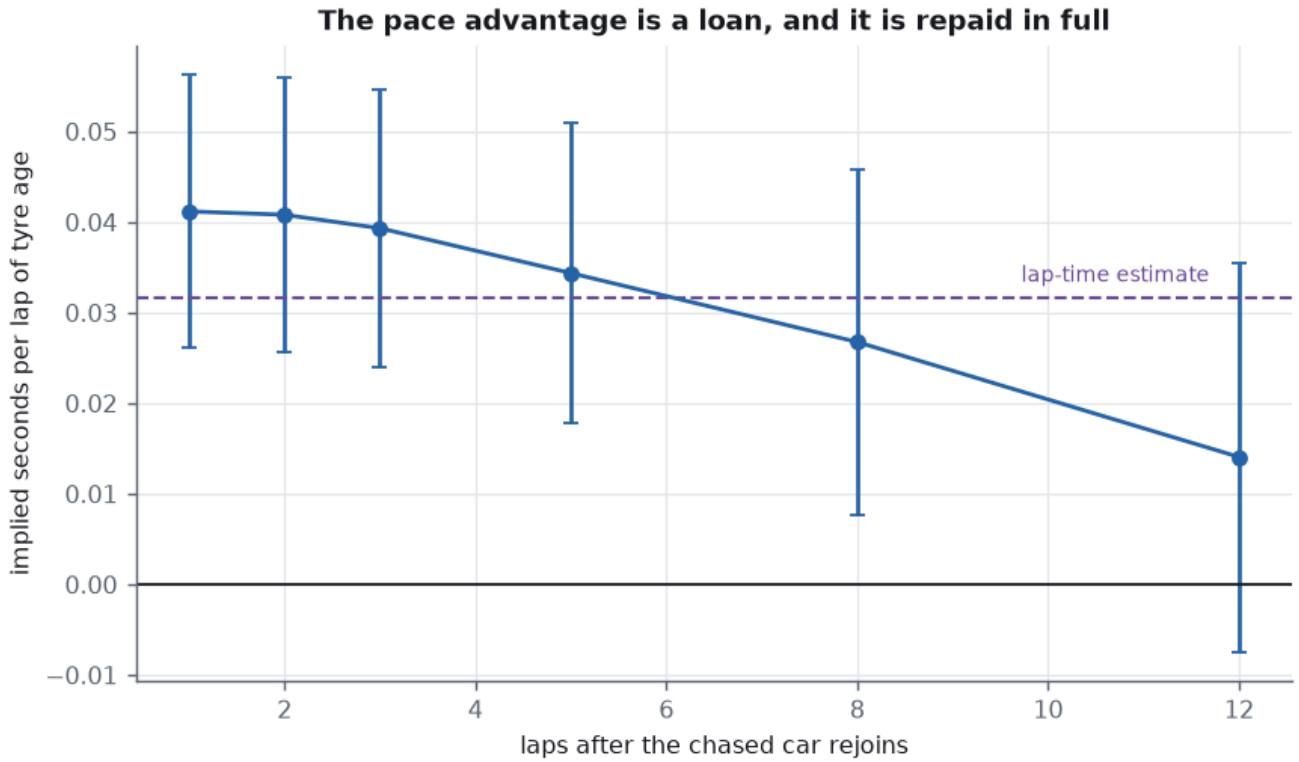
> The gap-based estimate is 1.30× the lap-time one, which is the direction a tyre cliff predicts — a linear fit through stints that ended before the tyre went away understates the penalty. But the intervals overlap, so this is agreement between two methods rather than a contradiction, and one season cannot resolve the difference.

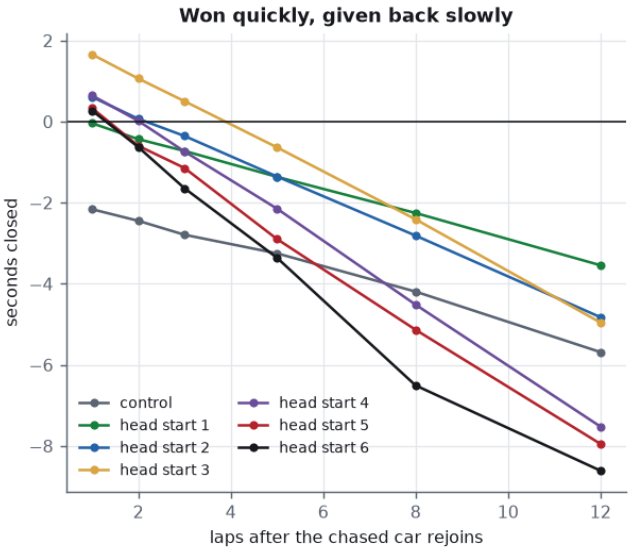


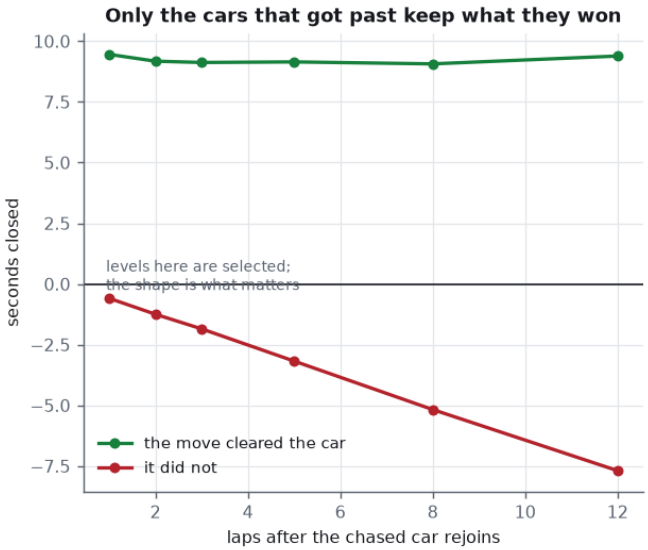
estimator	seconds per lap of age	95% interval
gaps between two cars (this project)	+0.0412	+0.0261 to +0.0562
lap times against tyre age (carried over)	+0.0316	+0.0256 to +0.0376

The pace advantage is a loan

Read later and later, the implied penalty decays to nothing. That is exactly the pure-pace prediction: once the rival has stopped, the chasing car is the one on the older set, and it hands the gain back at the same rate it won it. On pace alone, an undercut buys nothing permanent.





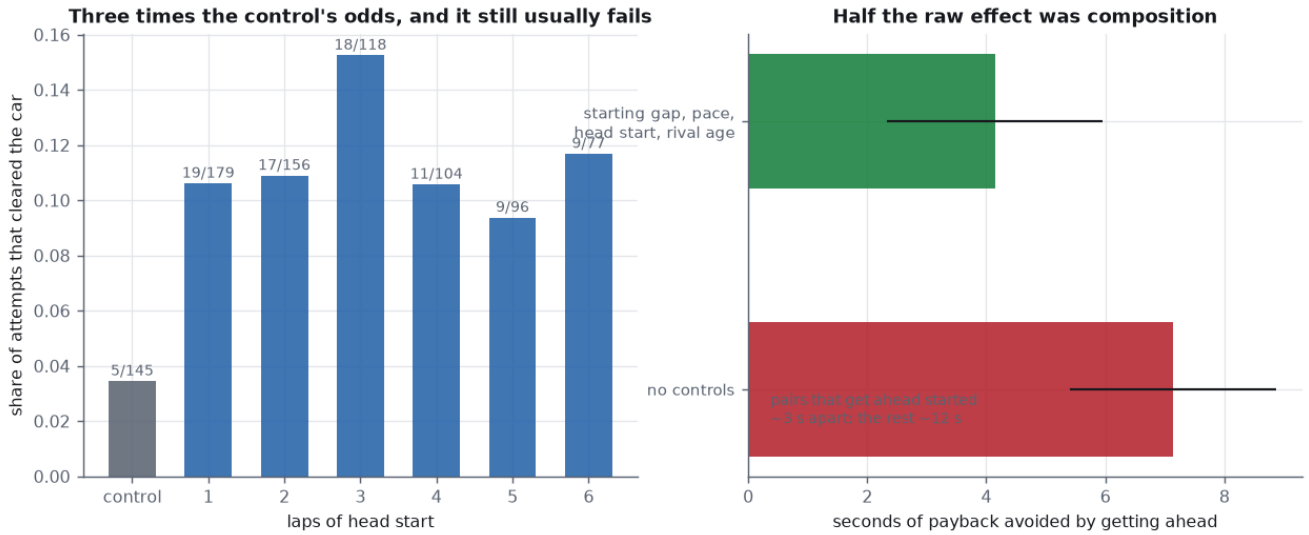


The pace advantage is a loan (continued)

laps after the rival rejoins	implied penalty	95% interval	pairs
1	+0.0412	+0.0261 to +0.0562	875
2	+0.0408	+0.0257 to +0.0559	875
3	+0.0393	+0.0239 to +0.0547	874
5	+0.0344	+0.0178 to +0.0509	871
8	+0.0267	+0.0077 to +0.0458	867
12	+0.0141	-0.0074 to +0.0355	857

But track position is not handed back

Split by whether the move worked, the two groups behave completely differently: the cars that got past hold what they won, and the ones that did not give it all back. The levels in this table are selected — a pair is in the first group because of its value at the early horizon — so only the shape is informative, and the shape is tested with controls below.



horizon	cleared the car	did not
1	+9.46	-0.59
2	+9.18	-1.25
3	+9.12	-1.85
5	+9.15	-3.18
8	+9.07	-5.17
12	+9.39	-7.69

What the controls do to it

Half the raw effect was composition. Without controls, getting ahead appears to be worth +7.14 s of avoided payback. But pairs that end up ahead started about three seconds apart and pairs that do not started about twelve, and a pair that starts close is a pair of cars of similar pace. With the starting gap in the model the figure is +4.14 s — real, highly significant, and about half what the raw comparison suggested.

term	estimate	95% interval	p
payback when still behind	-3.035	-4.923 to -1.148	0.0018
ahead after the move	+4.142	+2.338 to +5.947	0.0000
gap before	-0.325	-0.409 to -0.241	0.0000
prior pace	-0.931	-2.131 to +0.270	0.1278
head start	-0.860	-1.142 to -0.578	0.0000
rival age	+0.166	+0.076 to +0.256	0.0004

How often it works at all

Undercutting roughly triples the odds of getting past, from 3.4% for a same-lap stop to 11.4%. It still fails about 89% of the time, which is the part a time-trial model of strategy cannot tell you.

laps of head start	attempts	cleared	share	still ahead at 12 laps	held
control (same lap)	145	5	3.4%	5	100%
1	179	19	10.6%	15	79%
2	156	17	10.9%	13	76%
3	118	18	15.3%	15	83%
4	104	11	10.6%	9	82%
5	96	9	9.4%	7	78%
6	77	9	11.7%	5	56%

Lap times and pit stops from the Jolpica mirror of the Ergast schema. The cleaning rules and the lap-time estimator are carried over from the stint-degradation and race-strategy-mc projects.